

## QUALITATIVE AND QUANTITATIVE ANALYSIS FOR GENRE INDISTINGUISHABILITY

Tables I:(a) and II:(a) present pair-wise genre sample count for *fiction* and *non-fiction* genre in  $\mathcal{D}_{train}$ . Each cell in the table represents the number of samples belonging to both the row and column genre class IDs. Tables I:(b) and II:(b) illustrates the co-occurrence ratio between row class ID  $i$  in association with column class ID  $j$  for *fiction* and *non-fiction* genres, respectively. Each cell contains the ratio of samples belonging to both class IDs  $i$  and  $j$  to the total number of training samples associated with class ID  $j$ . Tables I:(c) and II:(c) reports the misclassification rate by IMAGINE for *fiction* and *non-fiction* genres, respectively. Here, each cell corresponding to row class ID  $i$  and column class ID  $j$  represents the ratio between the number of testing samples associated with class ID  $j$  but wrongly identified as class ID  $i$  and the total number of testing samples associated with class ID  $j$  that have been misclassified.

Our analysis of Tables I and II suggest that the high co-occurrence of the two genres, coupled with a significant imbalance in the number of training samples associated with one genre without the other, potentially causes the high

misclassification rate. For instance, in Table I, 264 out of 396 *press & media* (ID: 21) books are associated with *sci-fi & fantasy* (ID: 29) genre (refer to Table I:(a)). This leaves only 132 samples that are not categorized as *sci-fi & fantasy*. Conversely, there are 3716 *sci-fi & fantasy* books that do not belong to the *press & media* genre, creating an imbalance between non-*sci-fi & fantasy press & media* and non-*press & media sci-fi & fantasy* samples. This leads to *press & media* genre being misclassified as *sci-fi & fantasy* with a misclassification rate of 0.79. Standard data augmentation methods are limited in their ability to address this imbalance issue, as increasing the number of samples for *press & media* genre, also increases the *sci-fi & fantasy* samples due to inter-genre relation. We observe a similar issues between *humanities* (ID: 14) and *literature* (ID: 16), *animals & wildlife & pets* (ID: 1) and *childrens' book* (ID: 4) (refer to Table I), and *non-fiction* genres *teen & young adult* (ID: 27) and *history* (ID: 13) (refer to Table II). As a result, relatively lower  $\mathcal{BA}$  is observed for *press & media*, *humanities*, *animals & wildlife & pets* in *fiction*, and *teen & young adult* in *non-fiction*. Table III showcases the example of misclassified data samples due to genre indistinguishability.

TABLE I: Fiction genre indistinguishability analysis

| Class ID         | 1  | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   | 21   | 22   | 23   | 24   | 25   | 26   | 27   | 28   | 29   |      |   |
|------------------|----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|---|
| (a) Sample Count |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |   |
| Actual Class ID  | 1  | 1002 | 539  | 0    | 817  | 23   | 0    | 153  | 0    | 206  | 143  | 0    | 102  | 41   | 38   | 190  | 362  | 38   | 45   | 158  | 52   | 0    | 0    | 16   | 33   | 27   | 0    | 248  | 0    | 304  |   |
|                  | 2  | 539  | 2024 | 12   | 1279 | 32   | 68   | 308  | 44   | 206  | 319  | 24   | 141  | 129  | 54   | 435  | 716  | 46   | 81   | 204  | 137  | 23   | 102  | 154  | 44   | 71   | 53   | 327  | 39   | 882  |   |
|                  | 3  | 0    | 12   | 510  | 0    | 28   | 18   | 0    | 0    | 0    | 0    | 0    | 0    | 154  | 109  | 44   | 160  | 0    | 0    | 213  | 30   | 14   | 0    | 0    | 106  | 0    | 0    | 14   | 69   | 0    |   |
|                  | 4  | 817  | 1279 | 0    | 2575 | 104  | 69   | 291  | 44   | 262  | 387  | 0    | 174  | 273  | 72   | 449  | 997  | 70   | 109  | 484  | 242  | 14   | 51   | 74   | 25   | 83   | 189  | 979  | 19   | 876  |   |
|                  | 5  | 23   | 32   | 28   | 104  | 643  | 19   | 9    | 0    | 0    | 0    | 0    | 0    | 62   | 32   | 270  | 32   | 0    | 0    | 137  | 0    | 9    | 19   | 69   | 17   | 0    | 19   | 56   | 0    | 299  |   |
|                  | 6  | 0    | 68   | 18   | 69   | 19   | 473  | 0    | 0    | 0    | 0    | 0    | 0    | 39   | 18   | 54   | 137  | 0    | 0    | 154  | 0    | 0    | 19   | 16   | 18   | 0    | 179  | 16   | 36   | 419  |   |
|                  | 7  | 153  | 308  | 0    | 291  | 9    | 0    | 646  | 0    | 16   | 62   | 38   | 102  | 24   | 40   | 191  | 306  | 16   | 0    | 133  | 16   | 8    | 0    | 92   | 17   | 0    | 0    | 52   | 32   | 103  |   |
|                  | 8  | 0    | 44   | 0    | 44   | 0    | 0    | 0    | 408  | 0    | 0    | 77   | 0    | 103  | 0    | 0    | 191  | 0    | 0    | 268  | 44   | 0    | 52   | 0    | 52   | 0    | 0    | 52   | 0    | 0    |   |
|                  | 9  | 206  | 206  | 0    | 262  | 0    | 0    | 16   | 0    | 713  | 40   | 26   | 0    | 141  | 104  | 40   | 441  | 0    | 0    | 158  | 84   | 0    | 0    | 36   | 48   | 0    | 50   | 76   | 25   | 184  |   |
|                  | 10 | 143  | 319  | 0    | 387  | 0    | 0    | 62   | 0    | 40   | 638  | 80   | 48   | 130  | 0    | 103  | 371  | 31   | 0    | 20   | 48   | 0    | 0    | 97   | 0    | 36   | 0    | 94   | 8    | 28   |   |
|                  | 11 | 0    | 24   | 0    | 0    | 0    | 0    | 38   | 77   | 26   | 80   | 1128 | 96   | 254  | 139  | 99   | 614  | 89   | 54   | 256  | 151  | 0    | 9    | 376  | 0    | 153  | 0    | 18   | 66   | 159  |   |
|                  | 12 | 102  | 141  | 0    | 174  | 0    | 0    | 102  | 0    | 0    | 48   | 96   | 566  | 30   | 200  | 79   | 365  | 257  | 0    | 84   | 21   | 0    | 0    | 9    | 0    | 0    | 0    | 142  | 0    | 76   | 0 |
|                  | 13 | 41   | 129  | 154  | 273  | 62   | 39   | 24   | 103  | 141  | 130  | 254  | 30   | 2473 | 138  | 70   | 985  | 108  | 155  | 650  | 382  | 73   | 20   | 574  | 77   | 187  | 65   | 400  | 174  | 506  |   |
|                  | 14 | 38   | 54   | 109  | 72   | 32   | 18   | 40   | 0    | 104  | 0    | 139  | 200  | 138  | 1169 | 85   | 693  | 226  | 40   | 256  | 287  | 0    | 34   | 97   | 8    | 229  | 26   | 146  | 86   | 339  |   |
|                  | 15 | 190  | 435  | 44   | 449  | 270  | 54   | 191  | 0    | 40   | 103  | 99   | 79   | 70   | 85   | 1595 | 604  | 25   | 172  | 271  | 38   | 40   | 42   | 100  | 19   | 34   | 40   | 159  | 83   | 454  |   |
|                  | 16 | 362  | 716  | 160  | 997  | 32   | 137  | 306  | 191  | 441  | 371  | 614  | 365  | 985  | 693  | 604  | 4730 | 378  | 218  | 1044 | 496  | 67   | 216  | 803  | 219  | 277  | 88   | 899  | 425  | 1175 |   |
|                  | 17 | 158  | 204  | 213  | 484  | 137  | 154  | 133  | 268  | 158  | 20   | 256  | 84   | 650  | 256  | 271  | 1044 | 200  | 214  | 3371 | 258  | 168  | 0    | 379  | 158  | 90   | 236  | 471  | 242  | 1184 |   |
|                  | 18 | 38   | 46   | 0    | 70   | 0    | 0    | 16   | 0    | 0    | 31   | 89   | 257  | 108  | 226  | 25   | 378  | 752  | 0    | 200  | 29   | 46   | 0    | 117  | 24   | 0    | 8    | 114  | 0    | 144  |   |
|                  | 19 | 45   | 81   | 0    | 109  | 0    | 0    | 0    | 0    | 0    | 0    | 54   | 0    | 155  | 40   | 172  | 218  | 0    | 598  | 214  | 29   | 0    | 64   | 74   | 16   | 0    | 85   | 0    | 182  | 0    |   |
|                  | 20 | 52   | 137  | 30   | 242  | 0    | 0    | 16   | 44   | 84   | 48   | 151  | 21   | 382  | 287  | 38   | 496  | 29   | 258  | 1517 | 0    | 73   | 352  | 18   | 615  | 0    | 171  | 56   | 427  | 0    |   |
|                  | 21 | 0    | 23   | 14   | 14   | 9    | 0    | 8    | 0    | 0    | 0    | 0    | 0    | 73   | 0    | 40   | 67   | 46   | 0    | 168  | 0    | 396  | 23   | 0    | 16   | 0    | 0    | 14   | 264  | 0    |   |
|                  | 22 | 0    | 102  | 0    | 51   | 19   | 19   | 0    | 52   | 0    | 0    | 9    | 0    | 20   | 34   | 42   | 216  | 0    | 64   | 0    | 73   | 23   | 560  | 0    | 118  | 0    | 63   | 76   | 11   | 379  |   |
|                  | 23 | 16   | 154  | 0    | 74   | 69   | 16   | 92   | 0    | 36   | 97   | 376  | 9    | 574  | 97   | 100  | 803  | 117  | 74   | 379  | 352  | 0    | 0    | 1778 | 49   | 193  | 12   | 247  | 67   | 423  |   |
|                  | 24 | 33   | 44   | 0    | 25   | 17   | 18   | 17   | 52   | 48   | 0    | 0    | 0    | 77   | 8    | 19   | 219  | 24   | 16   | 158  | 18   | 16   | 118  | 49   | 580  | 0    | 92   | 0    | 281  | 0    |   |
|                  | 25 | 27   | 71   | 106  | 83   | 0    | 0    | 0    | 0    | 0    | 36   | 153  | 0    | 187  | 229  | 34   | 277  | 0    | 0    | 90   | 615  | 0    | 0    | 193  | 0    | 799  | 0    | 51   | 45   | 81   |   |
|                  | 26 | 0    | 53   | 0    | 189  | 19   | 179  | 0    | 0    | 50   | 0    | 0    | 0    | 65   | 26   | 40   | 88   | 8    | 0    | 236  | 0    | 0    | 63   | 12   | 0    | 0    | 657  | 114  | 0    | 366  |   |
|                  | 27 | 248  | 327  | 0    | 979  | 56   | 16   | 52   | 52   | 76   | 94   | 18   | 142  | 400  | 146  | 159  | 899  | 114  | 85   | 471  | 171  | 0    | 76   | 247  | 92   | 51   | 114  | 1973 | 0    | 766  |   |
|                  | 28 | 0    | 39   | 14   | 19   | 0    | 36   | 32   | 0    | 25   | 8    | 66   | 0    | 174  | 86   | 83   | 425  | 0    | 0    | 242  | 56   | 14   | 11   | 67   | 0    | 45   | 0    | 603  | 94   | 0    |   |
|                  | 29 | 304  | 882  | 69   | 876  | 299  | 419  | 103  | 0    | 184  | 28   | 159  | 76   | 506  | 339  | 454  | 1175 | 144  | 182  | 1184 | 427  | 264  | 379  | 423  | 281  | 81   | 366  | 766  | 94   | 3980 |   |
| (b) Cooccurrence |    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |   |
| Actual Class ID  | 1  | 1    | 0.27 | 0    | 0.32 | 0.04 | 0    | 0.24 | 0    | 0.29 | 0.22 | 0    | 0.18 | 0.02 | 0.03 | 0.12 | 0.08 | 0.05 | 0.08 | 0.05 | 0.03 | 0    | 0    | 0.01 | 0.06 | 0.03 | 0    | 0.13 | 0    | 0.08 |   |
|                  | 2  | 0.54 | 1    | 0.02 | 0.50 | 0.05 | 0.14 | 0.48 | 0.11 | 0.29 | 0.50 | 0.02 | 0.25 | 0.05 | 0.05 | 0.27 | 0.15 | 0.06 | 0.14 | 0.06 | 0.09 | 0.06 | 0.18 | 0.09 | 0.08 | 0.09 | 0.08 | 0.17 | 0.06 | 0.22 |   |
|                  | 3  | 0    | 0.01 | 1    | 0    | 0.04 | 0.04 | 0    | 0    | 0    | 0    | 0    | 0    | 0.06 | 0.09 | 0.03 | 0.03 | 0    | 0    | 0.06 | 0.02 | 0.04 | 0    | 0    | 0.13 | 0    | 0    | 0.02 | 0.02 | 0    |   |
|                  | 4  | 0.82 | 0.63 | 0    | 1    | 0.16 | 0.15 | 0.45 | 0.11 | 0.37 | 0.61 | 0    | 0.31 | 0.11 | 0.06 | 0.28 | 0.21 | 0.09 | 0.18 | 0.14 | 0.16 | 0.04 | 0.09 | 0.04 | 0.04 | 0.10 | 0.29 | 0.50 | 0.03 | 0.22 |   |
|                  | 5  | 0.02 | 0.02 | 0.05 | 0.04 | 1    | 0.04 | 0.01 | 0    | 0    | 0    | 0    | 0    | 0.03 | 0.03 | 0.17 | 0.01 | 0    | 0    | 0.04 | 0    | 0.02 | 0.03 | 0.04 | 0.03 | 0    | 0.03 | 0.03 | 0    | 0.08 |   |
|                  | 6  | 0    | 0.03 | 0.04 | 0.03 | 0.03 | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0.02 | 0.02 | 0.03 | 0.03 | 0    | 0    | 0.05 | 0    | 0    | 0.03 | 0.01 | 0.03 | 0    | 0.27 | 0.01 | 0.06 | 0.11 |   |
|                  | 7  | 0.15 | 0.15 | 0    | 0.11 | 0.01 | 0    | 1    | 0    | 0.02 | 0.10 | 0.03 | 0.18 | 0.01 | 0.03 | 0.12 | 0.06 | 0.02 | 0    | 0.04 | 0.01 | 0.02 | 0    | 0.05 | 0.03 | 0    | 0    | 0.03 | 0.05 | 0.03 |   |
|                  | 8  | 0    | 0.02 | 0    | 0.02 | 0    | 0    | 0    | 1    | 0    | 0    | 0.07 | 0    | 0.04 | 0    | 0    | 0.04 | 0    | 0    | 0.08 | 0.03 | 0    | 0.09 | 0    | 0.09 | 0    | 0    | 0.03 | 0    | 0    |   |
|                  | 9  | 0.21 | 0.10 | 0    | 0.10 | 0    | 0    | 0.02 | 0    | 1    | 0.06 | 0.02 | 0    | 0.06 | 0.09 | 0.03 | 0.09 | 0    | 0    | 0.05 | 0.06 | 0    | 0    | 0.02 | 0.08 | 0    | 0.08 | 0.04 | 0.04 | 0.05 |   |
|                  | 10 | 0.14 | 0.16 | 0    | 0.15 | 0    | 0    | 0.10 | 0    | 0.06 | 1    | 0.07 | 0.08 | 0.05 | 0    | 0.06 | 0.08 | 0.04 | 0    | 0.01 | 0.03 | 0    | 0    | 0.05 | 0    | 0.05 | 0    | 0.05 | 0.01 | 0.01 |   |
|                  | 11 | 0    | 0.01 | 0    | 0    | 0    | 0    | 0.06 | 0.19 | 0.04 | 0.13 | 1    | 0.17 | 0.10 | 0.12 | 0.06 | 0.13 | 0.12 | 0.09 | 0.08 | 0.10 | 0    | 0.02 | 0.21 | 0    | 0.19 | 0    | 0.01 | 0.11 | 0.04 |   |
|                  | 12 | 0.10 | 0.07 | 0    | 0.07 | 0    | 0    | 0.16 | 0    | 0    | 0.08 | 0.09 | 1    | 0.01 | 0.17 | 0.05 | 0.08 | 0.34 | 0    | 0.02 | 0.01 | 0    | 0    | 0.01 | 0    | 0    | 0    | 0.07 | 0    | 0.02 |   |
|                  | 13 | 0.04 | 0.06 | 0.30 | 0.11 | 0.10 | 0.08 | 0.04 | 0.25 | 0.20 | 0.20 | 0.23 | 0.05 | 1    | 0.12 | 0.04 | 0.21 | 0.14 | 0.26 | 0.19 | 0.25 | 0.18 | 0.04 | 0.32 | 0.13 | 0.23 | 0.10 | 0.20 | 0.29 | 0.13 |   |
|                  | 14 | 0.04 | 0.03 | 0.21 | 0.03 | 0.05 | 0.04 | 0.06 | 0    | 0.15 | 0    | 0.12 | 0.35 | 0.06 | 1    | 0.05 | 0.15 | 0.30 | 0.07 | 0.08 | 0.19 | 0    | 0.06 | 0.05 | 0.01 | 0.29 | 0.04 | 0.07 | 0.14 | 0.09 |   |
|                  | 15 | 0.19 | 0.21 | 0.09 | 0.17 | 0.42 | 0.11 | 0.30 | 0    | 0.06 | 0.16 | 0.09 | 0.14 | 0.03 | 0.07 | 1    | 0.13 | 0.03 | 0.29 | 0.08 | 0.03 | 0.10 | 0.08 | 0.06 | 0.03 | 0.04 | 0.06 | 0.08 | 0.14 | 0.11 |   |
|                  | 16 | 0.36 | 0.35 | 0.31 | 0.39 | 0.05 | 0.29 | 0.47 | 0.47 | 0.62 | 0.58 | 0.54 | 0.64 | 0.40 | 0.59 | 0.38 | 1    | 0.50 | 0.36 | 0.31 | 0.33 | 0.17 | 0.39 | 0.45 | 0.38 | 0.35 | 0.13 | 0.46 | 0.70 | 0.30 |   |
|                  | 17 | 0.16 | 0.10 | 0.42 | 0.19 | 0.21 | 0.33 | 0.21 | 0.66 | 0.22 | 0.03 | 0.23 | 0.15 | 0.26 | 0.22 | 0.17 | 0.22 | 0.27 | 0.36 | 1    | 0.17 | 0.42 | 0    | 0.21 | 0.27 | 0.11 | 0.36 | 0.24 | 0.40 | 0.30 |   |
|                  | 18 | 0.04 | 0.02 | 0    | 0.03 | 0    | 0    | 0.02 | 0    | 0    | 0.05 | 0.08 | 0.45 | 0.04 | 0.19 | 0.02 | 0.88 | 1    | 0    | 0.06 | 0.02 | 0.1  |      |      |      |      |      |      |      |      |   |

TABLE II: Non-fiction genre indistinguishability analysis

| Class ID         | 1                | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   | 21   | 22   | 23   | 24   | 25   | 26   | 27   | 28   | 30   |      |
|------------------|------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| (a) Sample Count |                  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Actual Class ID  | 1                | 748  | 169  | 0    | 222  | 41   | 0    | 12   | 0    | 306  | 14   | 0    | 12   | 89   | 52   | 97   | 44   | 14   | 10   | 0    | 16   | 0    | 95   | 0    | 369  | 14   | 0    | 65   | 37   | 157  |
|                  | 2                | 169  | 1444 | 9    | 331  | 203  | 42   | 0    | 143  | 53   | 34   | 103  | 12   | 512  | 237  | 165  | 109  | 26   | 57   | 58   | 58   | 270  | 408  | 125  | 236  | 14   | 8    | 162  | 28   | 259  |
|                  | 3                | 0    | 9    | 614  | 0    | 46   | 197  | 0    | 0    | 0    | 24   | 0    | 0    | 197  | 303  | 38   | 9    | 0    | 0    | 0    | 34   | 44   | 84   | 0    | 58   | 227  | 0    | 8    | 0    | 107  |
|                  | 4                | 222  | 331  | 0    | 878  | 41   | 0    | 8    | 28   | 84   | 170  | 0    | 31   | 214  | 140  | 82   | 52   | 37   | 46   | 0    | 9    | 0    | 152  | 0    | 340  | 61   | 9    | 315  | 0    | 109  |
|                  | 5                | 41   | 203  | 46   | 41   | 806  | 0    | 0    | 17   | 19   | 10   | 47   | 17   | 308  | 274  | 298  | 50   | 0    | 25   | 24   | 16   | 8    | 216  | 11   | 176  | 19   | 0    | 19   | 16   | 187  |
|                  | 6                | 0    | 42   | 197  | 0    | 0    | 483  | 0    | 0    | 11   | 0    | 0    | 0    | 165  | 183  | 11   | 0    | 8    | 12   | 0    | 0    | 0    | 69   | 0    | 201  | 28   | 9    | 0    | 0    | 73   |
|                  | 7                | 12   | 0    | 0    | 8    | 0    | 0    | 494  | 0    | 72   | 0    | 9    | 183  | 38   | 26   | 26   | 38   | 41   | 9    | 0    | 11   | 0    | 185  | 12   | 29   | 89   | 0    | 8    | 104  | 111  |
|                  | 8                | 0    | 143  | 0    | 28   | 17   | 0    | 0    | 439  | 45   | 0    | 110  | 8    | 52   | 44   | 8    | 19   | 0    | 34   | 0    | 32   | 0    | 290  | 0    | 0    | 47   | 8    | 0    | 0    | 12   |
|                  | 9                | 306  | 53   | 0    | 84   | 19   | 11   | 72   | 45   | 846  | 0    | 0    | 66   | 197  | 132  | 37   | 39   | 11   | 0    | 76   | 17   | 9    | 174  | 0    | 361  | 9    | 55   | 41   | 122  | 196  |
|                  | 10               | 14   | 34   | 24   | 170  | 10   | 0    | 0    | 0    | 889  | 119  | 142  | 16   | 521  | 33   | 25   | 149  | 84   | 0    | 193  | 0    | 222  | 225  | 54   | 506  | 9    | 80   | 0    | 0    | 57   |
|                  | 11               | 0    | 103  | 0    | 0    | 47   | 0    | 9    | 110  | 0    | 119  | 896  | 0    | 210  | 398  | 85   | 216  | 12   | 68   | 47   | 71   | 33   | 223  | 282  | 39   | 133  | 12   | 0    | 9    | 204  |
|                  | 12               | 12   | 12   | 0    | 31   | 17   | 0    | 183  | 8    | 66   | 142  | 0    | 949  | 17   | 381  | 0    | 11   | 456  | 0    | 0    | 92   | 0    | 400  | 11   | 221  | 506  | 16   | 0    | 0    | 84   |
|                  | 13               | 89   | 512  | 197  | 214  | 308  | 165  | 38   | 52   | 197  | 16   | 210  | 17   | 3297 | 1016 | 250  | 556  | 128  | 258  | 383  | 249  | 412  | 478  | 198  | 590  | 9    | 84   | 266  | 369  | 1198 |
|                  | 14               | 52   | 237  | 303  | 140  | 274  | 183  | 26   | 44   | 132  | 521  | 398  | 381  | 1016 | 3417 | 295  | 594  | 353  | 254  | 103  | 532  | 241  | 632  | 369  | 597  | 783  | 24   | 255  | 92   | 581  |
|                  | 15               | 97   | 165  | 38   | 82   | 298  | 11   | 26   | 8    | 37   | 33   | 85   | 0    | 250  | 295  | 1112 | 97   | 0    | 37   | 39   | 12   | 104  | 149  | 11   | 77   | 33   | 52   | 40   | 147  | 405  |
|                  | 16               | 44   | 109  | 9    | 52   | 50   | 0    | 38   | 19   | 39   | 25   | 216  | 11   | 556  | 594  | 97   | 1448 | 34   | 236  | 93   | 126  | 102  | 207  | 265  | 66   | 17   | 0    | 96   | 204  | 575  |
|                  | 17               | 0    | 58   | 0    | 0    | 24   | 0    | 0    | 0    | 76   | 0    | 47   | 0    | 383  | 103  | 39   | 93   | 44   | 41   | 744  | 55   | 126  | 103  | 87   | 58   | 9    | 81   | 0    | 221  | 383  |
|                  | 18               | 14   | 26   | 0    | 37   | 0    | 8    | 41   | 0    | 11   | 149  | 12   | 456  | 128  | 353  | 0    | 34   | 886  | 0    | 44   | 143  | 9    | 241  | 20   | 318  | 361  | 0    | 0    | 0    | 179  |
|                  | 19               | 10   | 57   | 0    | 46   | 25   | 12   | 9    | 34   | 0    | 84   | 68   | 0    | 258  | 254  | 37   | 236  | 0    | 762  | 41   | 52   | 0    | 315  | 0    | 32   | 27   | 0    | 25   | 37   | 289  |
|                  | 20               | 16   | 58   | 34   | 9    | 16   | 0    | 11   | 32   | 17   | 193  | 71   | 92   | 249  | 532  | 12   | 126  | 143  | 52   | 55   | 1122 | 0    | 135  | 205  | 93   | 416  | 12   | 0    | 36   | 147  |
|                  | 21               | 0    | 270  | 44   | 0    | 8    | 0    | 0    | 0    | 9    | 0    | 33   | 0    | 412  | 241  | 104  | 102  | 9    | 0    | 126  | 0    | 735  | 104  | 42   | 42   | 0    | 11   | 8    | 67   | 278  |
|                  | 22               | 95   | 408  | 84   | 152  | 216  | 69   | 185  | 290  | 174  | 222  | 223  | 400  | 478  | 632  | 149  | 207  | 241  | 315  | 103  | 135  | 104  | 2495 | 171  | 327  | 462  | 129  | 149  | 72   | 107  |
|                  | 23               | 0    | 125  | 0    | 0    | 11   | 0    | 12   | 0    | 0    | 225  | 282  | 11   | 198  | 369  | 11   | 265  | 20   | 0    | 87   | 205  | 42   | 171  | 877  | 70   | 273  | 12   | 0    | 43   | 162  |
|                  | 24               | 369  | 236  | 58   | 340  | 176  | 201  | 29   | 0    | 361  | 54   | 39   | 221  | 590  | 597  | 77   | 66   | 318  | 32   | 58   | 93   | 42   | 327  | 70   | 1809 | 148  | 0    | 188  | 41   | 207  |
|                  | 25               | 14   | 14   | 227  | 61   | 19   | 28   | 89   | 47   | 9    | 506  | 133  | 506  | 9    | 783  | 33   | 17   | 361  | 27   | 9    | 416  | 0    | 462  | 273  | 148  | 1519 | 25   | 16   | 9    | 63   |
|                  | 26               | 0    | 8    | 0    | 9    | 0    | 9    | 0    | 8    | 55   | 9    | 12   | 16   | 84   | 24   | 52   | 0    | 0    | 0    | 81   | 12   | 11   | 129  | 12   | 0    | 25   | 424  | 0    | 69   | 185  |
|                  | 27               | 65   | 162  | 8    | 315  | 19   | 0    | 8    | 0    | 41   | 80   | 0    | 0    | 266  | 255  | 40   | 96   | 0    | 25   | 0    | 0    | 8    | 149  | 0    | 188  | 16   | 0    | 659  | 0    | 151  |
|                  | 28               | 37   | 28   | 0    | 0    | 16   | 0    | 104  | 0    | 122  | 0    | 9    | 0    | 369  | 92   | 147  | 204  | 0    | 37   | 221  | 36   | 67   | 72   | 43   | 41   | 9    | 69   | 0    | 888  | 543  |
|                  | 30               | 157  | 259  | 107  | 109  | 187  | 73   | 111  | 12   | 196  | 57   | 204  | 84   | 1198 | 581  | 405  | 575  | 179  | 289  | 383  | 147  | 278  | 107  | 162  | 207  | 63   | 185  | 151  | 543  | 2802 |
|                  | (b) Cooccurrence |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Actual Class ID  | 1                | 1    | 0.12 | 0    | 0.25 | 0.05 | 0    | 0.02 | 0    | 0.36 | 0.02 | 0    | 0.01 | 0.03 | 0.02 | 0.09 | 0.03 | 0.02 | 0.01 | 0    | 0.01 | 0    | 0.04 | 0    | 0.20 | 0.01 | 0    | 0.10 | 0.04 | 0.06 |
|                  | 2                | 0.23 | 1    | 0.01 | 0.38 | 0.25 | 0.09 | 0    | 0.33 | 0.06 | 0.04 | 0.11 | 0.01 | 0.16 | 0.07 | 0.15 | 0.08 | 0.03 | 0.07 | 0.08 | 0.05 | 0.37 | 0.16 | 0.14 | 0.13 | 0.01 | 0.02 | 0.25 | 0.03 | 0.09 |
|                  | 3                | 0    | 0.01 | 1    | 0    | 0.06 | 0.41 | 0    | 0    | 0    | 0.03 | 0    | 0    | 0.06 | 0.09 | 0.03 | 0.01 | 0    | 0    | 0.03 | 0.06 | 0.03 | 0    | 0.03 | 0.15 | 0    | 0.01 | 0    | 0.04 | 0.04 |
|                  | 4                | 0.30 | 0.23 | 0    | 1    | 0.05 | 0    | 0.02 | 0.06 | 0.10 | 0.19 | 0    | 0.03 | 0.06 | 0.04 | 0.07 | 0.04 | 0.04 | 0.06 | 0    | 0.01 | 0    | 0.06 | 0    | 0.19 | 0.04 | 0.02 | 0.48 | 0    | 0.04 |
|                  | 5                | 0.05 | 0.14 | 0.07 | 0.05 | 1    | 0    | 0    | 0.04 | 0.02 | 0.01 | 0.05 | 0.02 | 0.09 | 0.08 | 0.27 | 0.03 | 0    | 0.03 | 0.03 | 0.01 | 0.01 | 0.09 | 0.01 | 0.10 | 0.01 | 0    | 0.03 | 0.02 | 0.07 |
|                  | 6                | 0    | 0.03 | 0.32 | 0    | 0    | 1    | 0    | 0    | 0.01 | 0    | 0    | 0    | 0.05 | 0.05 | 0.01 | 0    | 0.01 | 0.02 | 0    | 0    | 0.03 | 0    | 0.11 | 0.02 | 0.02 | 0    | 0    | 0    | 0.03 |
|                  | 7                | 0.02 | 0    | 0    | 0.01 | 0    | 0    | 1    | 0    | 0.09 | 0    | 0.01 | 0.19 | 0.01 | 0.01 | 0.02 | 0.03 | 0.05 | 0.01 | 0    | 0.01 | 0    | 0.07 | 0.01 | 0.02 | 0.06 | 0    | 0.01 | 0.12 | 0.04 |
|                  | 8                | 0    | 0.10 | 0    | 0.03 | 0.02 | 0    | 0    | 1    | 0.05 | 0    | 0.12 | 0.01 | 0.02 | 0.01 | 0.01 | 0.01 | 0    | 0.04 | 0    | 0.03 | 0    | 0.12 | 0    | 0    | 0.03 | 0.02 | 0    | 0    | 0    |
|                  | 9                | 0.41 | 0.04 | 0    | 0.10 | 0.02 | 0.02 | 0.15 | 0.10 | 1    | 0    | 0    | 0.07 | 0.06 | 0.04 | 0.03 | 0.03 | 0.01 | 0    | 0.10 | 0.02 | 0.01 | 0.07 | 0    | 0.20 | 0.01 | 0.13 | 0.06 | 0.14 | 0.07 |
|                  | 10               | 0.02 | 0.02 | 0.04 | 0.19 | 0.01 | 0    | 0    | 0    | 0    | 1    | 0.13 | 0.15 | 0    | 0.15 | 0.03 | 0.02 | 0.17 | 0.11 | 0    | 0.17 | 0    | 0.09 | 0.26 | 0.03 | 0.33 | 0.02 | 0.12 | 0    | 0.02 |
|                  | 11               | 0    | 0.07 | 0    | 0    | 0.06 | 0    | 0.02 | 0.25 | 0    | 0.13 | 1    | 0    | 0.06 | 0.12 | 0.08 | 0.15 | 0.01 | 0.09 | 0.06 | 0.06 | 0.04 | 0.09 | 0.32 | 0.02 | 0.09 | 0.03 | 0    | 0.01 | 0.07 |
|                  | 12               | 0.02 | 0.01 | 0    | 0.04 | 0.02 | 0    | 0.37 | 0.02 | 0.08 | 0.16 | 0    | 1    | 0.01 | 0.11 | 0    | 0.01 | 0.51 | 0    | 0    | 0.08 | 0    | 0.16 | 0.01 | 0.12 | 0.33 | 0.04 | 0    | 0    | 0.03 |
|                  | 13               | 0.12 | 0.35 | 0.32 | 0.24 | 0.38 | 0.34 | 0.08 | 0.12 | 0.23 | 0.02 | 0.23 | 0.02 | 1    | 0.30 | 0.22 | 0.38 | 0.14 | 0.34 | 0.51 | 0.22 | 0.56 | 0.19 | 0.23 | 0.33 | 0.01 | 0.20 | 0.40 | 0.42 | 0.43 |
|                  | 14               | 0.07 | 0.16 | 0.49 | 0.16 | 0.34 | 0.38 | 0.05 | 0.10 | 0.16 | 0.59 | 0.44 | 0.40 | 0.31 | 1    | 0.27 | 0.41 | 0.40 | 0.33 | 0.14 | 0.47 | 0.33 | 0.25 | 0.42 | 0.33 | 0.52 | 0.06 | 0.39 | 0.10 | 0.21 |
|                  | 15               | 0.13 | 0.11 | 0.06 | 0.09 | 0.37 | 0.02 | 0.05 | 0.02 | 0.04 | 0.04 | 0.09 | 0    | 0.08 | 0.09 | 1    | 0.07 | 0    | 0.05 | 0.05 | 0.01 | 0.14 | 0.06 | 0.01 | 0.04 | 0.02 | 0.12 | 0.06 | 0.17 | 0.14 |
|                  | 16               | 0.06 | 0.08 | 0.01 | 0.06 | 0.06 | 0    | 0.08 | 0.04 | 0.05 | 0.03 | 0.24 | 0.01 | 0.17 | 0.17 | 0.09 | 1    | 0.04 | 0.31 | 0.13 | 0.11 | 0.14 | 0.08 | 0.30 | 0.04 | 0.01 | 0    | 0.15 | 0.23 | 0.21 |
|                  | 17               | 0    | 0.04 | 0    | 0    | 0.03 | 0    | 0    | 0    | 0.09 | 0    | 0.05 | 0    | 0.12 | 0.03 | 0.04 | 0.06 | 0.05 | 0.05 | 1    | 0.05 | 0.17 | 0.04 | 0.10 | 0.03 | 0.01 | 0.19 | 0    | 0.25 | 0.14 |
|                  | 18               | 0.02 | 0.02 | 0    | 0.04 | 0    | 0.02 | 0.08 | 0    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |

TABLE III: Example of misclassified data samples due to genre indistinguishability

| (d) 9780671656010: The Adventures of Alyx; <b>Author:</b> Joanna Russ; <b>Publisher:</b> Baen               |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Blurb           | Contents: Bluestocking (1967) I Thought She Was Afeard till She Stroked My Beard (1967) The Barbarian (1968) Picnic on Paradise (1968) novel The Second Inquisition (1970)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                             | Actual Genre    | (Fiction, (Humanities, <a href="#">Sci-Fi &amp; Fantasy</a> ))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                             | Predicted Genre | (Fiction, (Humanities, <a href="#">Literature</a> ))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (e) 9780445409132: Murder in the Queen's Armes; <b>Author:</b> Aaron Elkins; <b>Publisher:</b> Warner Books |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                            | Blurb           | Physical anthropologist Gideon Oliver is honeymooning in rural England when a side trip to an archaeological site leads to an adventure in scholarly intrigue and murder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                             | Actual Genre    | (Fiction, ( <a href="#">Humanities</a> , Literature, Mystery & Thriller & Suspense & Horror & Adventure))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                             | Predicted Genre | (Fiction, ( <a href="#">History</a> , Literature, Mystery & Thriller & Suspense & Horror & Adventure))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| (f) 9780789425225: Voices in the Park; <b>Author:</b> Anthony Browne; <b>Publisher:</b> DK Children         |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                          | Blurb           | Four people enter a park and through their eyes young readers see different visions, from the bossy woman and the sad man to the lonely boy and the warm young girl, moving from one voice to another and shifting landscapes and seasons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                                             | Actual Genre    | (Fiction, ( <a href="#">Animals &amp; Wildlife &amp; Pets</a> , Arts & Photography, Childrens' Book, Family & Parenting & Relationships, Teen & Young Adult))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                                             | Predicted Genre | (Fiction, (Arts & Photography, Childrens' Book, Family & Parenting & Relationships, Teen & Young Adult))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| (g) 9780440436836: Homesick: My Own Story; <b>Author:</b> Jean Fritz; <b>Publisher:</b> Yearling            |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                          | Blurb           | Jean Fritz was born in China and lived there until 1927, when she was twelve. Young Jean had spent her entire life in China, but her parents' memories of home and letters from relatives in Pennsylvania made her feel that she was American—and homesick for a place she'd never seen! Family photographs and illustrations by Margot Tones show us the real people behind Jean's vivid and unforgettable stories—memories of picnics on the Great Wall, pranks, holidays in the foreign compound, rebellious moments at her British school, close ties to Chinese friends, and how it felt to be called a "foreign devil" and spat upon in the streets of a turbulent China on the eve of revolution. When her family embarks upon its long journey home, Jean is thrilled, but she When she arrives in America at last, will she fit in after growing up on "the wrong side of the world?" |
|                                                                                                             | Actual Genre    | (Non-fiction, (Biographies & Memoir, Childrens' Book, History, <a href="#">Teen &amp; Young Adult</a> ))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                             | Predicted Genre | (Non-fiction, (Biographies & Memoir, Childrens' Book, History))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (h) 9780345291363: Elephant Man; <b>Author:</b> Christine Sparks; <b>Publisher:</b> Ballantine Books        |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                          | Blurb           | John Merrick had lived for more than twenty years imprisoned in a body that condemned him to a miserable life in the workhouse and to humiliation as a circus sideshow freak. But beneath that tragic exterior, within that enormous and deformed head, thrived the soul of a poet, the heart of a dreamer, the longings of a man. Based on the extraordinary motion picture that captured the heart of America.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                             | Actual Genre    | (Non-fiction, (Biographies & Memoir, History, Literature, <a href="#">Teen &amp; Young Adult</a> ))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                             | Predicted Genre | (Non-fiction, (Biographies & Memoir, History, Literature))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

False positives are marked in red, and False negatives are marked in blue